KU LEUVEN
ARENBERG DOCTORAL SCHOOL

SCIENCE, ENGINEERING & TECHNOLOGY
ARENBERG DOCTORAL SCHOOL
KASTEELPARK ARENBERG 11 BUS 2001
3001 HEVERLEE

Annual Progress Presentation

Name of the PhD researcher:	Alex Arokia Raj
Student Number:	0474317
Title of the PhD:	Investigation of Compression modes in nuclei far from stability
Name of the corresponding supervisor:	Riccardo Raabe
Date of the presentation	21.11.2024

PhD researchers need to report on the progress of their PhD on an annual basis by giving a presentation to their Supervisory Committee. This presentation should include the following elements:

- A report of activities already carried out (literature review, formulation of research objectives and methodology, experiments and results, publications, doctoral training diary,…);
- A detailed timing for the completion of the PhD;

Through this presentation and the following discussion, the Supervisory Committee verifies if the PhD research has made sufficient progress and with a feasible time schedule for the rest of the PhD, so that it can indeed be expected that he/she will finish the PhD programme in a reasonable time frame.

The PhD researcher and Supervisory Committee agree on the following time frame for the completion of the PhD: mention month and year when you plan to do this.

Submission of the doctoral training diary is foreseen: …April 2025……

Submission of the composition of the Examination Committee is foreseen: April 2025…

The preliminary defence is foreseen: …May 2025…

The public defence is foreseen: …June 2025…….

Based on these elements and a thorough discussion with the PhD researcher, the Supervisory Committee makes the following recommendations:

Alex's PhD work concerns the measurement of the isoscalar giant resonances in the nucleus 58Ni. The experiment was performed in the summer of 2019 at the GANIL accelerator facility (France). The analysis of the data is quite complex, and discrepancies are found with respect to a parallel work of colleagues in France. For this reason, a publication based on the results of the analysis seems premature. On the other hand, Alex has developed a novel method for the identification of particle tracks in a gaseous detector, which he has reported in an article, now under revision with the supervisors.

(continues on next page)

In his presentation before the supervisory committee, Alex has confirmed that he has fulfilled all the milestones of his PhD diary, except for a published article and the thesis; for the latter, the technical chapters have already been drafted.
The supervisory committee sets the following objectives for Alex's progress:
- the article should be finalised and sent to collaborators by the end of 2024;
- during the refereeing process of the article, Alex should concentrate on writing the introductory chapter(s) of the thesis -- indicatively by March at the latest -- about the physics background and motivation of his research work.
The quality of those chapters will be then evaluated by the supervisors to assess the prospects for the defence of the thesis.

The PhD researcher needs to upload this form, the presentation and the signed signature sheet via his/her KULoket tool.
The members of the Supervisory Committee must approve via email to the PhD researcher or by signing the signatures sheet which the PhD researcher downloaded via KULoket or by signing below:

Names and signatures of the members of the Supervisory Committee:

Riccardo Raabe Superviosor	Marine Vandebroeck External co-supervisor
Piet Van Duppen Assessor	Thomas Cocolios Assessor

From: **VANDEBROUCK Marine** marine.vandebrouck@cea.fr
Subject: RE: Report of Alex's presentation
Date: 26 November 2024 at 12:35
To: Thomas Elias Cocolios thomas.cocolios@kuleuven.be, Piet Van Duppen piet.vanduppen@kuleuven.be, Riccardo Raabe riccardo.raabe@kuleuven.be

VM

Dear Riccardo,

It is ok also for me.

All the best,
Marine

De : Thomas Elias Cocolios <thomas.cocolios@kuleuven.be>
Envoyé : mardi 26 novembre 2024 11:56:34
À : Piet Van Duppen; Riccardo Raabe; VANDEBROUCK Marine
Objet : Re: Report of Alex's presentation

Dear Riccardo,
I agree with this report.
Best regards,
Thomas..

My working day may not be your working day. Please do not feel obliged to reply to this email outside of your normal working hours.

Prof Thomas Elias Cocolios (he/him)
thomas.cocolios@kuleuven.be
KU Leuven
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Institute for Nuclear & Radiation Physics
Interdisciplinary Research Group
https://fys.kuleuven.be/iks/interdisciplinary-research
tel: +32 16 32 31 16
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From: Piet Van Duppen <piet.vanduppen@kuleuven.be>
Date: Tuesday, 26 November 2024 at 10:42
To: Riccardo Raabe <riccardo.raabe@kuleuven.be>, VANDEBROUCK Marine <marine.vandebrouck@cea.fr>, Thomas Elias Cocolios <thomas.cocolios@kuleuven.be>
Subject: Re: Report of Alex's presentation

Dear Riccardo,

I agree with this report!

Take care,
Piet

Piet Van Duppen

KU Leuven
Instituut voor Kern- en Stralingsfysica

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Website: www.fys.kuleuven.be/iks/
www.kuleuven.be/wieiswie/en/person/00010286

From: Riccardo Raabe <riccardo.raabe@kuleuven.be>
Sent: Tuesday, November 26, 2024 10:30
To: VANDEBROUCK Marine <marine.vandebrouck@cea.fr>; Thomas Elias Cocolios <thomas.cocolios@kuleuven.be>; Piet Van Duppen <piet.vanduppen@kuleuven.be>
Subject: Report of Alex's presentation

Dear all,
Here is a short report of the meeting on Alex's progress.
If you agree just reply, or you may suggest some changes of course.
Thanks
Riccardo

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In his presentation before the supervisory committee, Alex has confirmed that he has fulfilled all the milestones of his PhD diary, except for a published article and the thesis; for the latter, the technical chapters have already been drafted.
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